



Beginner Friendly ML Learning Resources

Data Cleaning, EDA, Feature Engineering & ML Pipelines

All resources listed here are **free to access**. YouTube videos are free. Articles on GeeksforGeeks, Towards Data Science, Analytics Vidhya, and Machine Learning Mastery are free. Kaggle notebooks are free with a free account.



Best Websites for Data Science Students

Website	Purpose	Difficulty
Kaggle Learn	Hands-on learning	Beginner
Seeing Theory	Statistics visualization	Beginner
StatQuest	Statistics + ML	Beginner
Google ML Crash Course	Machine learning	Beginner
Distill Archive	Deep learning visualization	Intermediate
TensorFlow Playground	Neural networks	Beginner
Scikit-learn User Guide	ML implementation	Intermediate
Machine Learning Mastery	Practical ML	Beginner
OpenIntro Statistics	Statistics textbook	Beginner
Think Stats	Statistics with Python	Beginner
Statistics by Jim	Statistics	Beginner

Seeing Theory	Probably the most beautiful statistics learning website ever built.	Beginner
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Best Reference Books for Machine Learning

1. **"Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow"** By Aurélien Géron
2. **"Machine Learning with Python Cookbook"** By Chris Albon
3. **"Introduction to Machine Learning with Python"** By Andreas C. Müller and Sarah Guido
4. **"Feature Engineering for Machine Learning"** By Alice Zheng
5. **"Practical Statistics for Data Scientists"** By Peter Bruce and Andrew Bruce
6. **"Think Stats"** By Allen B. Downey



Recommended YouTube Channels (Watch These All Year!)

Channel	Best For	Link
StatQuest with Josh Starmer	Statistics & ML concepts with fun visuals	https://www.youtube.com/@statquest

Krish Naik	End-to-end ML projects, Python, practical tutorials	https://www.youtube.com/@krishnaik06
CampusX	Hindi/Urdu-friendly beginner ML & Python tutorials	https://www.youtube.com/@campusx-official
Data School (Kevin Markham)	Pandas, Scikit-learn, data cleaning — very beginner-friendly	https://www.youtube.com/@dataschool
Corey Schafer	Python basics, Pandas, Matplotlib, NumPy	https://www.youtube.com/@coreyms
freeCodeCamp.org	Full-length free bootcamp-style courses	https://www.youtube.com/@freecodecamp
codebasics	Practical DS & ML tutorials in simple English	https://www.youtube.com/@codebasics



Interactive Probability Websites

1. Probability Explorer

Interactive probability experiments.

[Probability Explorer](#)

2. Bayes Theorem Visualizer

Makes Bayes theorem intuitive.

[Bayes Theorem Calculator and Visualizer](#)

3. Desmos Graphing Calculator

Excellent for statistics and distributions.



Data Cleaning & Preprocessing



Understanding Different Data Types

Articles & Blogs

- **GeeksforGeeks — Data Types in Python** <https://www.geeksforgeeks.org/python-data-types/> *Simple explanation of int, float, string, bool, object, datetime — great starting point.*
- **Pandas Official Docs — Data Types (dtype)** https://pandas.pydata.org/docs/user_guide/basics.html#dtypes *Learn how Pandas stores data internally — numeric, categorical, datetime, object.*
- **DataCamp — Understanding Data Types for Machine Learning** <https://www.datacamp.com/tutorial/guide-to-data-cleaning-in-python> *Covers numerical vs categorical types, why they matter in ML.*

YouTube Videos

- **Corey Schafer — Python Data Types** <https://www.youtube.com/watch?v=khKv-8q7YmY> *Clear, beginner-friendly walkthrough of Python types.*
- **codebasics — Pandas Data Types Tutorial** Search: `codebasics pandas data types` on YouTube *Practical notebook-style tutorial with dtypes, astype().*



Identifying & Handling Missing Values

Articles & Blogs

- **Apify Blog — Data Cleaning in Python (Beginner's Guide 2025)** <https://blog.apify.com/data-cleaning-python/> *Covers isnull(), fillna() with mean/median/mode, dropna() — with code examples.*
- **GeeksforGeeks — Handling Missing Values in Pandas** <https://www.geeksforgeeks.org/working-with-missing-data-in-pandas/> *Step-by-step: detecting, visualizing, and filling missing data.*

- **Medium — Data Cleaning & Preprocessing in Python (Khush Kumari, 2026)**
<https://medium.com/@khush.khushboo516/data-cleaning-preprocessing-in-python-f6cbfdb39f27> *Fresh article with isnull().sum(), mean/median imputation, and forward fill.*
- **DataQuest — Complete Guide to Data Cleaning in Python**
<https://www.dataquest.io/guide/data-cleaning-in-python-tutorial/> *Practical examples with real datasets.*

YouTube Videos

- **Krish Naik — Handling Missing Values (Complete Tutorial)** Search: **Krish Naik handling missing values pandas** on YouTube *Covers mean, median, mode, and forward/backward fill with clear examples.*
- **codebasics — Pandas Missing Values** Search: **codebasics pandas missing values tutorial** on YouTube
- **Corey Schafer — Pandas Tutorial (Part 6 — Cleaning Data)**
<https://www.youtube.com/watch?v=ZyhVh-qRZPA> *Real-world CSV cleaning with missing value handling.*

Outlier Detection (IQR & Z-Score)

Articles & Blogs

- **GeeksforGeeks — Data Preprocessing in Python (IQR section)**
<https://www.geeksforgeeks.org/data-analysis/data-preprocessing-machine-learning-python/> *Has working IQR code with NumPy percentile for outlier detection.*
- **Analytics Vidhya — Outlier Detection and Treatment**
<https://www.analyticsvidhya.com/blog/2021/05/detecting-and-treating-outliers-treating-the-odd-one-out/> *Excellent beginner guide covering both IQR and Z-score with visualizations.*
- **Towards Data Science — Ways to Detect and Remove Outliers**
<https://towardsdatascience.com/ways-to-detect-and-remove-the-outliers-404d16608dba/> *Visual and statistical methods explained clearly.*

YouTube Videos

- **StatQuest — Outliers and How to Handle Them** Search: **StatQuest outliers** on YouTube *Intuitive visual explanation of what an outlier is and why it matters.*

- **Krish Naik — Outlier Detection IQR Z-Score Search:** [Krish Naik outlier detection IQR Z-score](#) on YouTube *Step-by-step Python implementation.*

Outlier Treatment (Capping & Removal)

Articles & Blogs

- **Analytics Vidhya — Treating Outliers (Capping/Winsorization)**
[https://www.analyticsvidhya.com/blog/2021/05/detecting-and-treating-outliers-treating-the-odd-one-out/Covers-both-removal-and-capping-\(Winsorization\)-methods/](https://www.analyticsvidhya.com/blog/2021/05/detecting-and-treating-outliers-treating-the-odd-one-out/Covers-both-removal-and-capping-(Winsorization)-methods/)
- **GeeksforGeeks — Outlier Removal Python**
<https://www.geeksforgeeks.org/machine-learning/removing-outliers-using-iqr-in-pandas-dataframe/Simple-removal-using-boolean-masking/>

Data Type Conversion

Articles & Blogs

- **GeeksforGeeks — Pandas astype() Method**
<https://www.geeksforgeeks.org/python-pandas-dataframe-astype/Convert-between-int.-float.-str.-category,-datetime.>
- **Pandas Docs — Type Casting**
<https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.astype.html>

Duplicate Removal

Articles & Blogs

- **GeeksforGeeks — Removing Duplicates in Pandas**
[https://www.geeksforgeeks.org/python-pandas-dataframe-drop_duplicates/Covers-duplicated\(\),-drop_duplicates\(\),-and-subset-parameter.](https://www.geeksforgeeks.org/python-pandas-dataframe-drop_duplicates/Covers-duplicated(),-drop_duplicates(),-and-subset-parameter.)
- **DataCamp — Data Cleaning Guide (Duplicates Section)**
<https://www.datacamp.com/tutorial/guide-to-data-cleaning-in-python>

String Operations for Text Cleaning

Articles & Blogs

- **Medium — Data Cleaning & Preprocessing (String ops section, 2026)**
<https://medium.com/@khush.khushboo516/data-cleaning-preprocessing-in-python-f6cbfdb39f27> *Covers `.str.lower()`, `.str.strip()`, `.str.replace()`, `.str.title()`.*
- **GeeksforGeeks — Pandas String Methods**
<https://www.geeksforgeeks.org/python-pandas-series-str-lower-upper-and-title/>
- **Pandas Docs — Working with Text Data** https://pandas.pydata.org/docs/user_guide/text.html *Official documentation for all `.str.` methods — bookmark this!*

YouTube Videos

- **Corey Schafer — Pandas String Methods** Search: **Corey Schafer pandas string methods** on YouTube

EDA & Visualization

EDA Methodology

Articles & Blogs

- **GeeksforGeeks — EDA in Python (Full Guide)**
<https://www.geeksforgeeks.org/data-analysis/exploratory-data-analysis-in-python/> *Full methodology: inspection → missing values → univariate → bivariate → multivariate.*
- **GeeksforGeeks — EDA with NumPy, Pandas, Matplotlib & Seaborn**
<https://www.geeksforgeeks.org/data-analysis/eda-with-numpy-pandas-matplotlib-seaborn/> *End-to-end workflow with code.*
- **Medium — Data Visualization & EDA**
<https://medium.com/@lomashbhuvva/data-visualization-exploratory-data-analysis-eda-1c347fdd7505> *Explains when and why to use different visualization types.*

YouTube Videos

- **CampusX — Complete EDA Tutorial in Python** Search: **CampusX EDA tutorial Python** on YouTube *Very thorough, beginner-friendly, Hindi-English mixed.*
- **Krish Naik — Exploratory Data Analysis** Search: **Krish Naik EDA complete tutorial** on YouTube

Topic 2: Statistical Summaries (describe, info)

Articles & Blogs

- **Pandas Docs — describe() method**
<https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.describe.html>
- **GeeksforGeeks — Pandas describe()**
<https://www.geeksforgeeks.org/python-pandas-dataframe-describe-method/>

Distribution Analysis

Articles & Blogs

- **GeeksforGeeks — EDA (Distribution Section)**
<https://www.geeksforgeeks.org/data-analysis/exploratory-data-analysis-in-python/ Histograms. KDE plots, violin plots — all with code.>
- **Medium — EDA with Matplotlib & Seaborn (Practical Guide)**
<https://medium.com/artificial-intelligence-101/exploratory-data-analysis-eda-with-matplotlib-and-seaborn-a-practical-guide-9284672534df>

YouTube Videos

- **StatQuest — Histograms Explained** Search: **StatQuest histograms** on YouTube

Correlation Analysis

Articles & Blogs

- **GeeksforGeeks — Data Preprocessing — Correlation Section**
<https://www.geeksforgeeks.org/data-analysis/data-preprocessing-machine-learning-python/ Includes heatmap code with seaborn.>
- **StatQuest — Pearson's Correlation Article** Search on StatQuest website: <https://statquest.org>

YouTube Videos

- **StatQuest — Correlation Coefficient Explained** Search: **StatQuest correlation** on YouTube
Simple visual explanation of positive, negative, and no correlation.

Matplotlib Basics (Line, Bar, Scatter Plots)

Articles & Blogs

- **Medium — Visualizing Data Like a Pro: Matplotlib, Seaborn & Plotly**
<https://medium.com/@aditib259/visualizing-data-like-a-pro-matplotlib-seaborn-plotly-7a5001ab5514>
Beginner-friendly walkthrough with code for all basic chart types.
- **Matplotlib Official Tutorials** <https://matplotlib.org/stable/tutorials/index.html> *Official docs with interactive examples.*
- **W3Schools — Matplotlib Tutorial** https://www.w3schools.com/python/matplotlib_intro.asp *Super beginner-friendly. interactive code.*

YouTube Videos

- **Corey Schafer — Matplotlib Tutorial (Playlist)**
https://www.youtube.com/playlist?list=PL-osiE80TeTvipOqomVEeZ1HRrcEvtZB_ *Best Matplotlib playlist for beginners — 9 videos covering everything.*
- **freeCodeCamp — Matplotlib Full Course** Search: **freeCodeCamp matplotlib full course** on YouTube

Seaborn for Statistical Plots

Articles & Blogs

- **Seaborn Official Tutorial** <https://seaborn.pydata.org/tutorial.html> *Official tutorial with beautiful examples — bookmark this!*
- **GeeksforGeeks — Seaborn Tutorial** <https://www.geeksforgeeks.org/seaborn-tutorial/> *Covers all major plot types with code.*
- **Medium — EDA with Matplotlib and Seaborn**
<https://medium.com/artificial-intelligence-101/exploratory-data-analysis-eda-with-matplotlib-and-seaborn-a-practical-guide-9284672534df>

YouTube Videos

- **Krish Naik — Seaborn Full Tutorial** Search: `Krish Naik seaborn tutorial beginner` on YouTube
- **codebasics — Seaborn Tutorial for Beginners** Search: `codebasics seaborn tutorial` on YouTube

Plotly for Interactive Visualizations

Articles & Blogs

- **Plotly Official Docs — Getting Started** <https://plotly.com/python/getting-started/> *Official guide with interactive examples.*
- **Medium — Visualizing Data Like a Pro (Plotly Section)** <https://medium.com/@aditib259/visualizing-data-like-a-pro-matplotlib-seaborn-plotly-7a5001ab5514> *Explains how Plotly differs from Matplotlib — hover, zoom, pan.*
- **Medium — EDA Visualization with Plotly** <https://python.plainenglish.io/data-visualization-for-exploratory-data-analysis-eda-in-python-b6a4a09d6124>

YouTube Videos

- **Krish Naik — Plotly Tutorial for Beginners** Search: `Krish Naik plotly express tutorial` on YouTube
- **codebasics — Plotly Tutorial** Search: `codebasics plotly tutorial` on YouTube

Feature Scaling (StandardScaler, MinMaxScaler, RobustScaler)

Articles & Blogs

- **DataCamp — Normalization vs Standardization** <https://www.datacamp.com/tutorial/normalization-vs-standardization> *Clear explanation of when to use each scaler with Python code.*
- **Apify — Data Cleaning Python (Scaling Section)** <https://blog.apify.com/data-cleaning-python/MinMaxScaler-and-StandardScaler-with-scikit-learn-examples>
- **Scikit-learn Docs — Preprocessing** <https://scikit-learn.org/stable/modules/preprocessing.html> *Official reference for all three scalers.*
- **Analytics Vidhya — Feature Scaling for ML** <https://www.analyticsvidhya.com/blog/2020/04/feature-scaling-machine-learning-normalization-standard>

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YouTube Videos

- **Krish Naik — Feature Scaling StandardScaler MinMaxScaler** Search: **Krish Naik feature scaling standardscaler minmaxscaler** on YouTube
- **StatQuest — Feature Scaling** Search: **StatQuest feature scaling normalization** on YouTube

Normalization Techniques

Articles & Blogs

- **GeeksforGeeks — Normalization vs Standardization**
<https://www.geeksforgeeks.org/normalization-vs-standardization/>
- **Towards Data Science — When to Normalize vs Standardize**
<https://towardsdatascience.com/normalization-vs-standardization-which-one-is-better-f29e043a57eb/>

Encoding Categorical Variables

Articles & Blogs

- **DataCamp — Data Cleaning Guide (Encoding Section)**
<https://www.datacamp.com/tutorial/guide-to-data-cleaning-in-python> *Covers Label Encoding and One-Hot Encoding side by side.*
- **Analytics Vidhya — Encoding Categorical Variables**
<https://www.analyticsvidhya.com/blog/2020/08/types-of-categorical-data-encoding/> *Best article comparing Label, One-Hot, Ordinal, and Binary encoding.*
- **GeeksforGeeks — Label Encoding vs One Hot Encoding**
<https://www.geeksforgeeks.org/ml-label-encoding-of-datasets-in-python/>

YouTube Videos

- **Krish Naik — Label Encoding vs One Hot Encoding** Search: **Krish Naik label encoding one hot encoding** on YouTube *Explains when to use which — must watch!*
- **StatQuest — One Hot Encoding** Search: **StatQuest one hot encoding** on YouTube

Feature Creation and Extraction

Articles & Blogs

- **DataCamp — Feature Engineering**
<https://www.datacamp.com/tutorial/feature-engineering-deep-learning> *Practical guide to creating interaction features, polynomial features.*
- **Analytics Vidhya — Feature Engineering for ML**
<https://www.analyticsvidhya.com/blog/2018/08/guide-automated-feature-engineering-featuretools-python/>

Topic 5: Binning and Discretization

Articles & Blogs

- **Apify — Data Cleaning Python (Binning Section)** <https://blog.apify.com/data-cleaning-python/>
Explains `pd.cut()` and `pd.qcut()` with examples.
- **GeeksforGeeks — Pandas `cut()` and `qcut()`**
<https://www.geeksforgeeks.org/pandas-cut-method-in-python/>

Handling Date-Time Features

Articles & Blogs

- **GeeksforGeeks — Pandas DateTime Features**
<https://www.geeksforgeeks.org/python-working-with-date-and-time-using-pandas/>
- **Medium — Date-Time Feature Engineering**
<https://medium.com/@khush.khushboo516/data-cleaning-preprocessing-in-python-f6cbfdb39f27>
Converting strings to datetime objects, extracting year/month/day.

Train-Test Split & Data Leakage Prevention

Importance of Data Splitting

Articles & Blogs

- **Analytics Vidhya — Train Test Validation Split Explained**
<https://www.analyticsvidhya.com/blog/2020/08/train-test-split/>
- **Machine Learning Mastery — How to Split Data into Train and Test Sets**
<https://machinelearningmastery.com/train-test-split-for-evaluating-machine-learning-algorithms/>

YouTube Videos

- **StatQuest — Training, Testing, and Validation Sets Explained** Search: **StatQuest training testing validation** on YouTube *Visual and conceptual — explains WHY we split, not just how.*

Stratified Sampling

Articles & Blogs

- **GeeksforGeeks — Stratified Sampling in scikit-learn**
<https://www.geeksforgeeks.org/stratified-sampling-in-scikit-learn/> *Explains why class distribution must be preserved in splits.*

Cross-Validation (K-Fold, Stratified K-Fold)

Articles & Blogs

- **Analytics Vidhya — Cross Validation Tutorial**
<https://www.analyticsvidhya.com/blog/2018/05/improve-model-performance-cross-validation-in-python-r/>
Excellent beginner guide with diagrams explaining K-Fold.
- **Machine Learning Mastery — K-Fold Cross Validation**
<https://machinelearningmastery.com/k-fold-cross-validation/>
- **Scikit-learn Docs — Cross Validation** https://scikit-learn.org/stable/modules/cross_validation.html

YouTube Videos

- **StatQuest — Cross Validation Explained** Search: [StatQuest cross validation](#) on YouTube
Best visual explanation — must watch before coding it.
- **Krish Naik — K-Fold and Stratified K-Fold** Search: [Krish Naik K-fold cross validation Python](#) on YouTube

Data Leakage — Concepts & Prevention

Articles & Blogs

- **Towards Data Science — The Right Way of Using SMOTE with Cross-Validation**
<https://towardsdatascience.com/the-right-way-of-using-smote-with-cross-validation-92a8d09d00c7/>
Great real-world example of leakage + how pipelines fix it.
- **Machine Learning Mastery — Data Leakage in Machine Learning**
<https://machinelearningmastery.com/data-leakage-machine-learning/> *Clear definition with examples of how leakage inflates metrics.*
- **Analytics Vidhya — Data Leakage and How to Avoid It**
<https://www.analyticsvidhya.com/blog/2021/07/data-leakage-in-machine-learning-a-guide-for-beginners/>

YouTube Videos

- **Krish Naik — Data Leakage in Machine Learning** Search: [Krish Naik data leakage machine learning](#) on YouTube

Handling Imbalanced Datasets

Articles & Blogs

- **Machine Learning Mastery — SMOTE for Imbalanced Classification**
<https://machinelearningmastery.com/smote-oversampling-for-imbalanced-classification/> *The definitive beginner guide to SMOTE with Python code.*
- **Analytics Vidhya — SMOTE Techniques**
<https://www.analyticsvidhya.com/blog/2020/10/overcoming-class-imbalance-using-smote-techniques/>
Covers SMOTE, ADASYN, and SMOTETomek.
- **GeeksforGeeks — SMOTE in Python**
<https://www.geeksforgeeks.org/machine-learning/smote-for-imbalanced-classification-with-python/>
Step-by-step implementation with credit card fraud dataset.

- **Train in Data Blog — SMOTE in Python (2025 Guide)**
<https://www.blog.trainindata.com/smote-in-python-a-guide-to-balanced-datasets/> *Very detailed, explains when and when NOT to use SMOTE.*

YouTube Videos

- **Krish Naik — Handling Imbalanced Dataset SMOTE** Search: **Krish Naik SMOTE imbalanced dataset** on YouTube
- **codebasics — Imbalanced Dataset Handling** Search: **codebasics imbalanced dataset SMOTE** on YouTube

Pipeline Creation for Preprocessing

Articles & Blogs

- **Scikit-learn Docs — Pipeline**
<https://scikit-learn.org/stable/modules/generated/sklearn.pipeline.Pipeline.html> *Official reference — essential reading.*
- **Towards Data Science — A Simple Guide to Scikit-learn Pipelines**
<https://towardsdatascience.com/a-simple-example-of-pipeline-in-machine-learning-with-scikit-learn-e726ffb6976/> *Beginner-friendly explanation with ColumnTransformer.*
- **Analytics Vidhya — Building ML Pipeline with scikit-learn**
<https://www.analyticsvidhya.com/blog/2020/01/build-your-first-machine-learning-pipeline-using-scikit-learn/>

YouTube Videos

- **Krish Naik — Scikit-Learn Pipeline Tutorial** Search: **Krish Naik sklearn pipeline tutorial** on YouTube
- **Data School — Machine Learning Pipelines with scikit-learn** Search: **Data School sklearn pipeline** on YouTube